

Activity 3 — Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies

Course Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)

WSQ Course Code TGS-2024049781

Topic Topic 01: Foundations of Design Thinking, Agile, and Generative AI for Problem-Solving

Duration 40 minutes · Group size: 3–5 participants

Tools ChatGPT or Microsoft Copilot, Padlet Classroom Board

Learning Outcome Addressed

LO1 · K1 — Latest trends in design thinking; how generative AI enhances design thinking and agile processes.

The Real Case

A Singapore bank's innovation team used a large language model to generate customer personas for a new savings product aimed at gig-economy workers. The AI produced five polished, confident personas in under a minute, complete with names, incomes, quotes and frustrations. The team built a concept around 'Marcus, 29, a Grab driver saving for a HDB flat'. When they finally ran six real interviews, the actual gig workers' dominant anxiety was not saving for a flat at all — it was irregular income smoothing within a single month, and a deep distrust of anything that locked their money away. The AI personas were plausible, internally consistent, well-written, and wrong. They were a compression of what the internet says about gig workers, not evidence about these gig workers.

Your Scenario

Your team has two weeks to produce a validated problem statement. A colleague proposes skipping the interviews because 'the AI personas are good enough and we're behind schedule'.

What You Will Do

Learners run the same persona-generation task through a GenAI tool, then interrogate the output against an evidence test, to build a working rule for where GenAI belongs in the design thinking loop and where its confident fluency becomes a liability.

What you will produce

A GenAI-generated persona set annotated with evidence gaps, and a group 'AI use rule' posted to Padlet.

Step-by-Step Instructions

1. Open ChatGPT or Microsoft Copilot.
2. Prompt: 'Create 3 detailed user personas for a savings product aimed at gig-economy workers in Singapore. Include name, age, income pattern, goals, frustrations and a representative quote.'
3. Read the output as a group and mark every claim that is an assumption rather than evidence.
4. Prompt the AI again: 'For each persona, list what evidence would be required to confirm it, and what would falsify it.'
5. Compare the two outputs — note how much more useful the second framing is.
6. Draft your group's one-sentence rule for using GenAI in discovery and post it to the Padlet Activity 3 section.
7. Like the rule you find most defensible from another group and comment why.

Discussion Questions

1. The AI personas were plausible but wrong. What property of large language models makes fluent output a poor proxy for truth?
2. Which parts of the design thinking loop can GenAI genuinely accelerate, and which parts must stay human? Draw the line and justify it.
3. Your colleague says the interviews can be skipped. Give the strongest version of their argument, then rebut it.
4. How would you make an AI-generated persona falsifiable — what evidence would prove it wrong?
5. What governance would you put in place so AI-generated insight is always labelled as unvalidated?

How You Know You Are Done

Your annotated persona set clearly separates assumption from evidence, and your posted rule states both where GenAI is allowed and where it is not.

Debrief — What the Room Should Conclude

A language model generates the statistically likely, not the locally true. That is exactly what you want for divergence — volume, range, provocations, first drafts — and exactly what you must not trust for convergence, where a wrong-but-confident persona sends a whole quarter's work in the wrong direction. The workable rule most groups arrive at: use GenAI to widen the funnel and to synthesise evidence you already collected; never use it as the evidence itself. Practically, every AI-generated artefact should carry a provenance label — who or what produced it, and what real-world evidence has since confirmed or refuted it. Note also the failure was caught only because someone eventually talked to six real people; six interviews is a small price for not building the wrong product.

